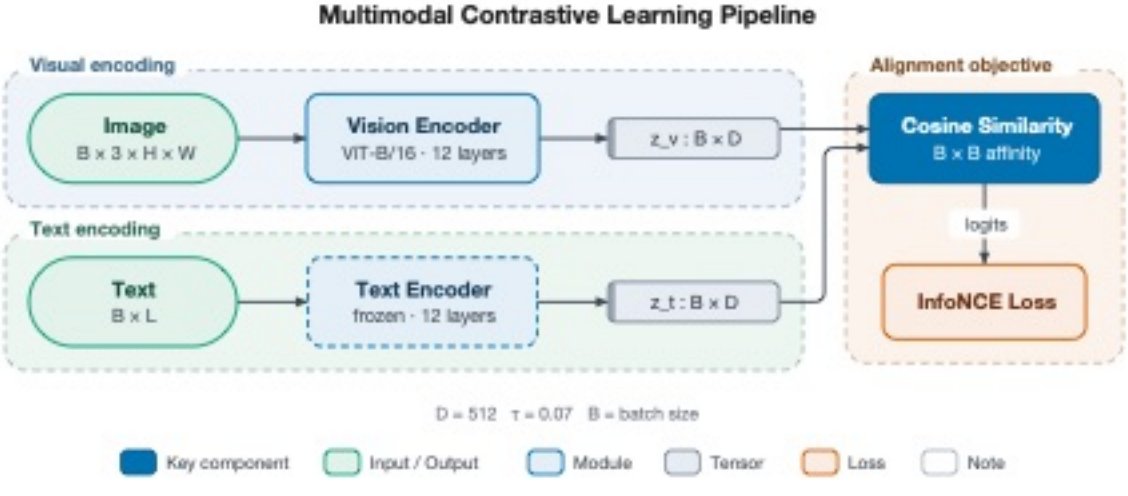
Multimodal Contrastive Learning Pipeline
Visual encoding
Image
B × 3 × H × W
Vision Encoder
ViT-B/16 · 12 layers
z_v : B × D
Text encoding
Text
B × L
Text Encoder
frozen · 12 layers
z_t : B × D
Alignment objective
Cosine Similarity
B × B affinity
logits
InfoNCE Loss
D = 512 τ = 0.07 B = batch size
Key component
Input / Output
Module
Tensor
Loss
Note